Sahaj Bhadja

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Summary

Software engineer with 2+ years building **C++/Python systems**, high-volume data and simulation pipelines, computer-vision applications, and tested cross-platform software; processed 650K+ records, improved planning 18%, and delivered production workflows with 99%+ reliability.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built **high-volume Python data services** and normalized storage schemas for 500+ users, optimizing indexed retrieval and access patterns to reduce application latency from 1.2 seconds to 350 milliseconds.
- Developed **fault-tolerant acquisition and processing pipelines** handling 5,000+ monthly jobs with incremental loads, validation, reconciliation, explicit states, failure isolation, and CI/CD releases, achieving 99%+ production reliability.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Engineered a **concurrent Python processing system** using multithreading and fault-tolerant execution, processing 1,000+ files per batch and 10,000+ daily requests while maintaining stable performance under peak production workloads.
- Integrated **multimodal vision and OCR models** through switchable APIs, adding validation, usage logging, fallback handling, and persistent results to reduce processing turnaround 60% while improving operational diagnosability.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built **simulation and data pipelines** processing 100K+ autonomous-system trajectories, integrating HD-map geometry and temporal features through multi-head attention to improve long-horizon planning performance 18% across controlled experiments.
- Designed **reproducible closed-loop evaluations** across three planning architectures, analyzing performance constraints and failure patterns to improve simulation stability 15%, reduce off-road trajectories, and strengthen map-aware control behavior.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Designed **object-oriented processing components** using Singleton and Factory patterns, AWS Glue, S3, SQL, and caching, improving response times 30%, pipeline efficiency 68%, storage costs 15%, and maintainability across distributed workflows.
- Automated **cross-platform builds, deployments, and 20+ tests** with GitHub, Jenkins, PyTest, and Mockito, generating coverage and failure reports while supporting reliable software releases across Windows, Linux, and macOS.

Cyber Security and Data Mining **India**
Data Analyst - Research Fellowship *May 2023 – August 2023*
- Built a **high-volume classification pipeline** over 650K+ records, engineering 21 statistical and behavioral features while comparing Random Forest, Gaussian Naive Bayes, RNN, LSTM, GRU, and CNN architectures.
- Achieved **96%+ detection accuracy** through reproducible model evaluation and feature analysis, identifying malformed and adversarial patterns while translating benchmark results into clear technical findings for collaborative research review.

Skills

Languages: C++, Python, Java, SQL, Bash, JavaScript, TypeScript
Systems: Linux, OOP, data structures, algorithms, multithreading, concurrency, performance optimization, simulation
Data: PostgreSQL, MySQL, SQL Server, MongoDB, AWS Glue, S3, ETL, indexing, storage optimization
ML & Vision: PyTorch, TensorFlow, Transformers, CNNs, computer vision, feature engineering, model evaluation
Quality: Git, Jenkins, GitHub Actions, CI/CD, PyTest, Mockito, Docker, unit/integration testing, Agile/Scrum

Projects

CPU Scheduling Simulator Live | *C++, React.js, Algorithms*
- Implemented **six C++ scheduling and optimization algorithms** with interactive controls, Gantt visualization, and dynamic performance metrics, testing 1,000+ process combinations while reducing average turnaround time 25% across benchmark scenarios.

PiXMod Live | *PyTorch, Transformers, SAM, MongoDB*
- Built an **image-processing and segmentation pipeline** using SAM, Stable Diffusion, and PyTorch, processing 1,000+ edited assets while indexing images and metadata in MongoDB for scalable storage and retrieval.

CaptionCraft Live | *Python, PyTorch, Transformers, Flask*
- Benchmarked **four transformer vision architectures** across 10K images and 100K captions, improving accuracy from 81% to 96% while building Flask inference APIs for scalable real-time caption generation.